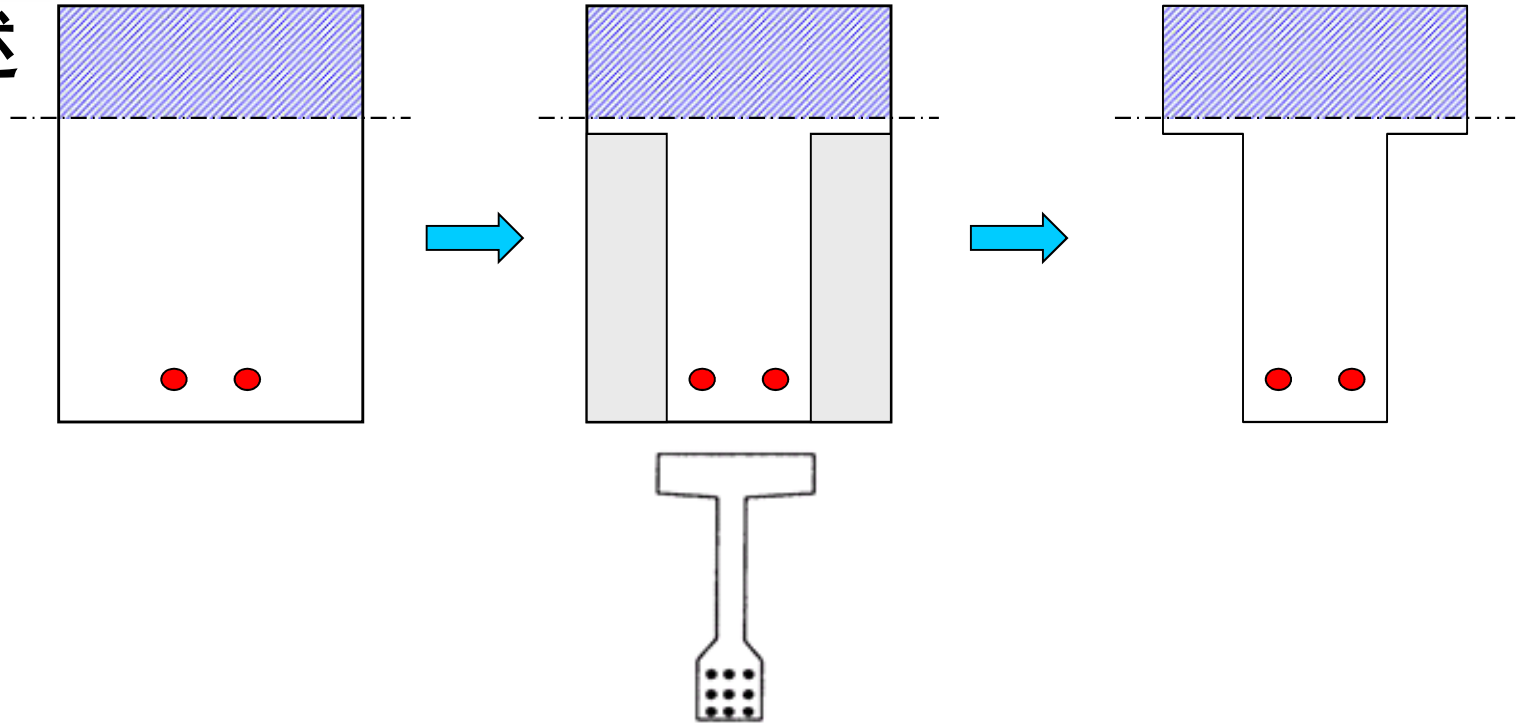
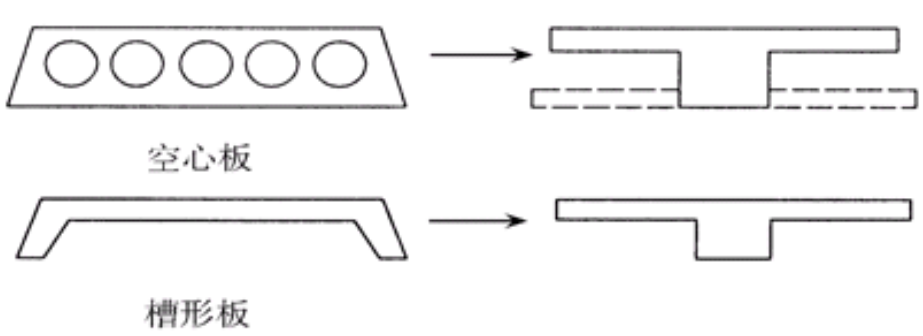
5.5 T形及I形截面梁

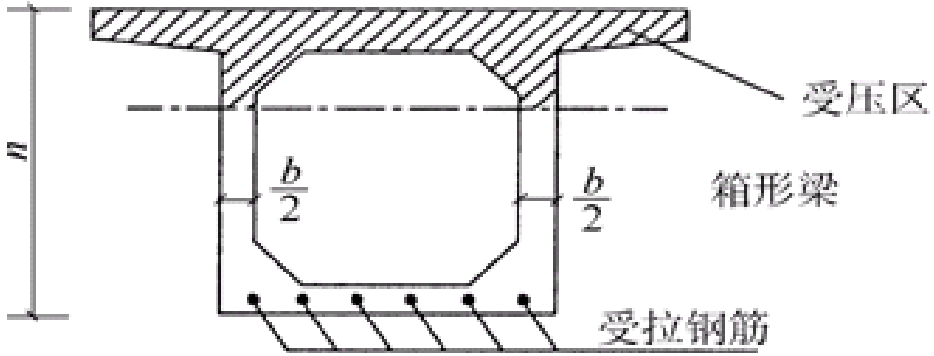
1. 概述
2. 两类T形截面的判断
3. 第一类T形截面
4. 第二类T形截面

5.5 T形及I形截面梁

1. 概述

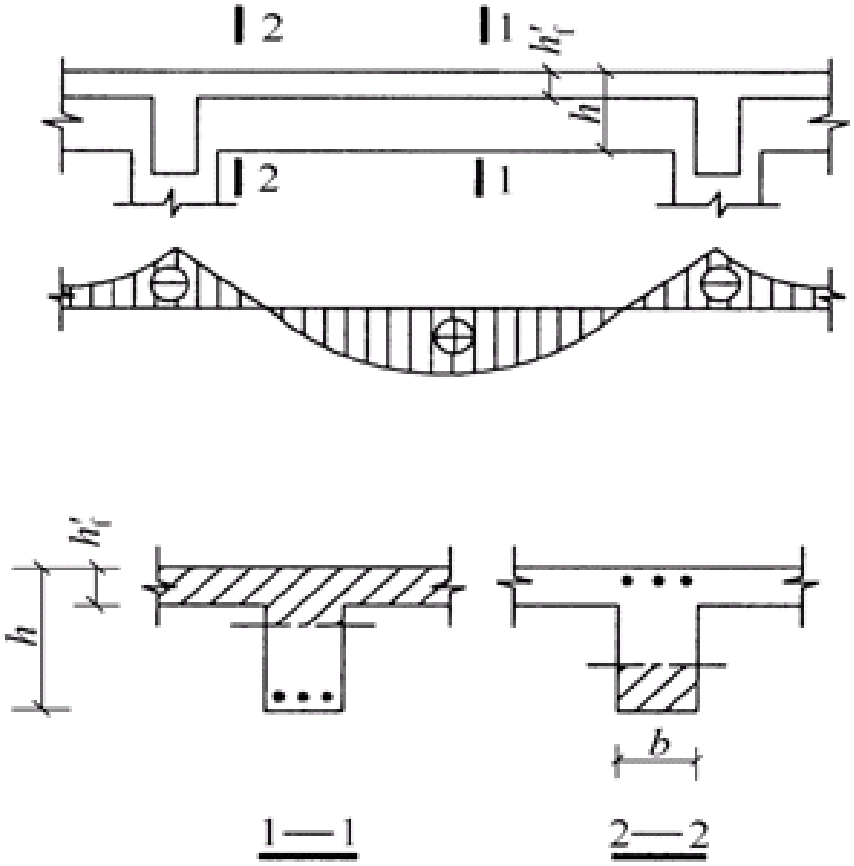


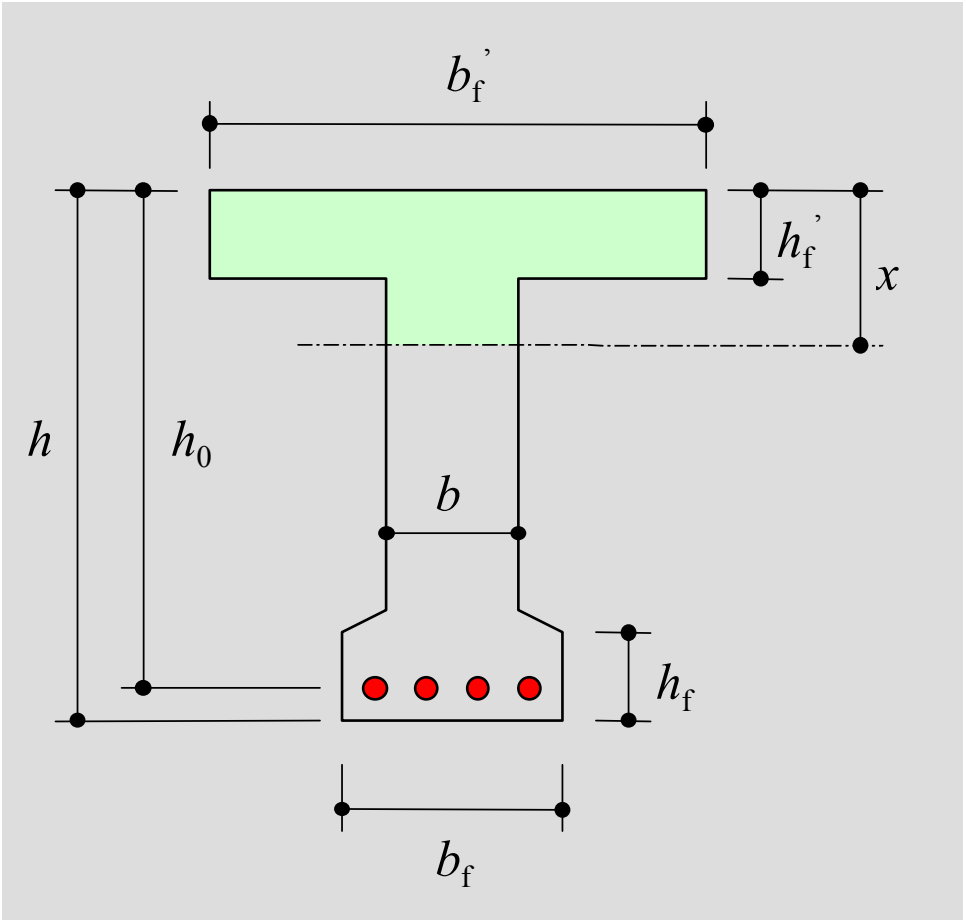




5.5 T形及I形截面梁

1. 概述

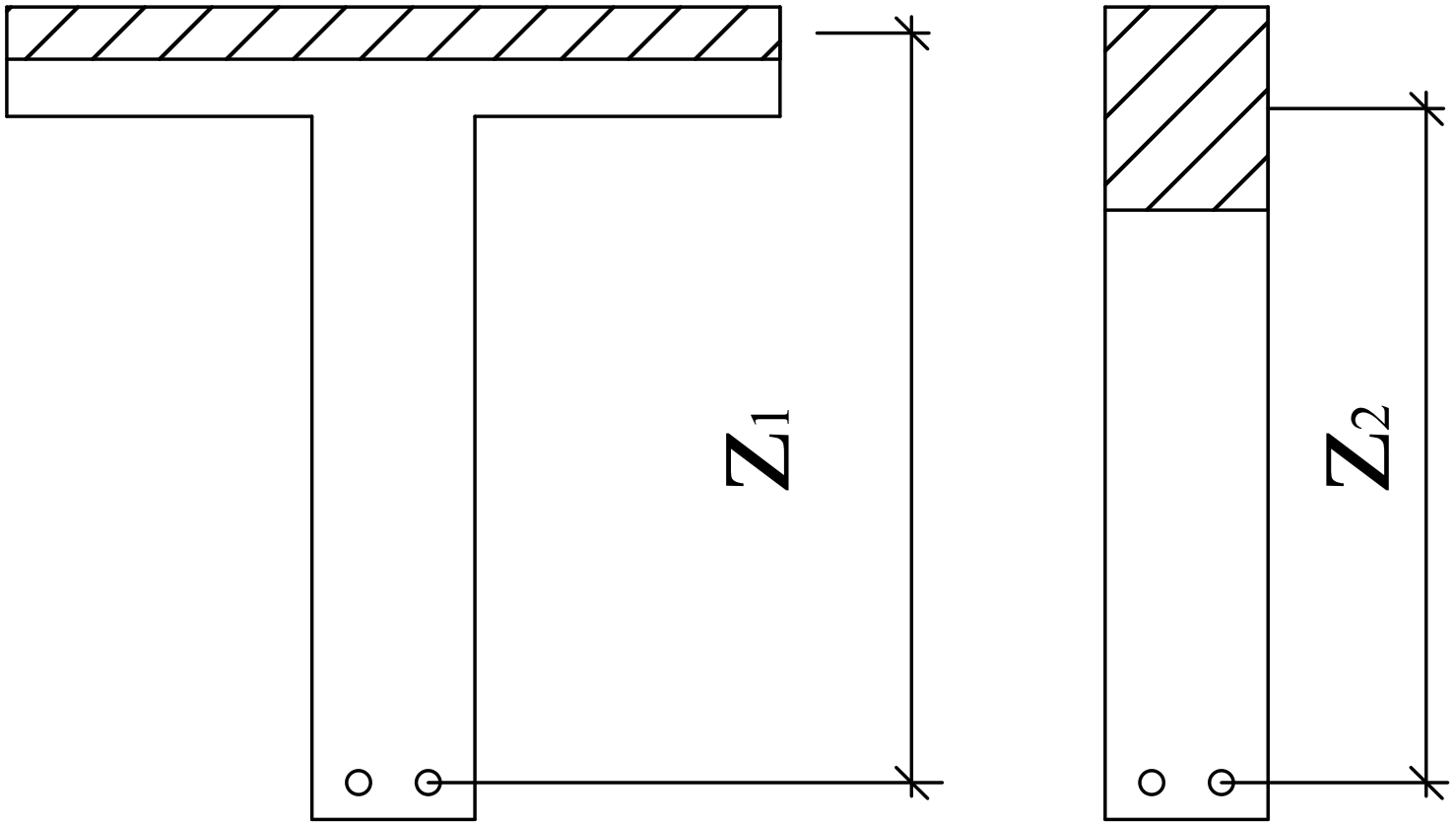




5.5 T形及I形截面梁

1. 概述

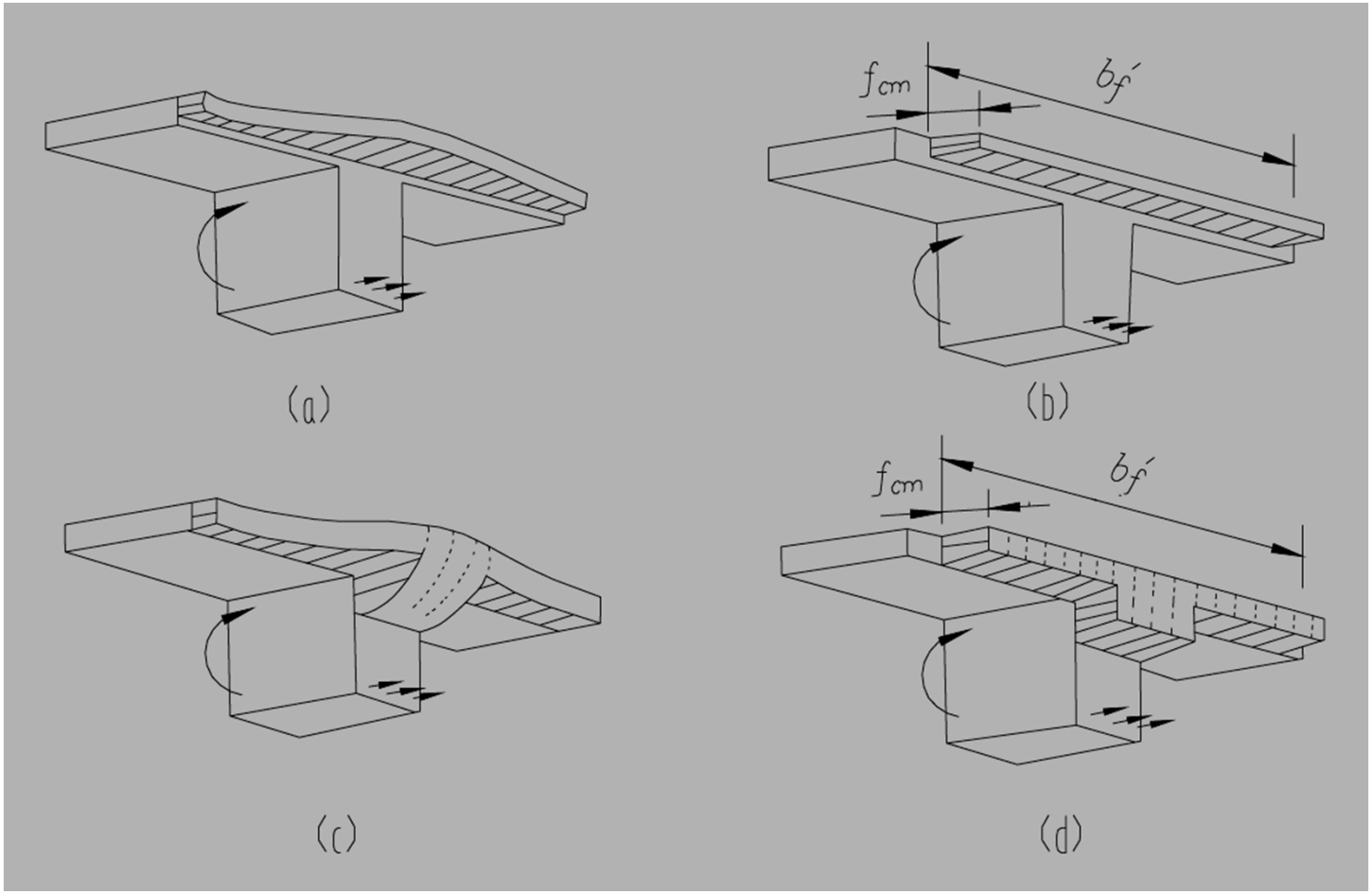
T形截面比矩形截面抗弯性能更好



5.5 T形及I形截面梁

1. 概述

T形截面受压翼缘的计算宽度

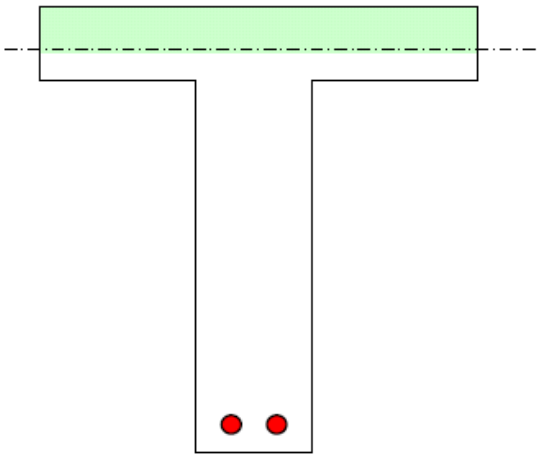


具体取值见表5.4

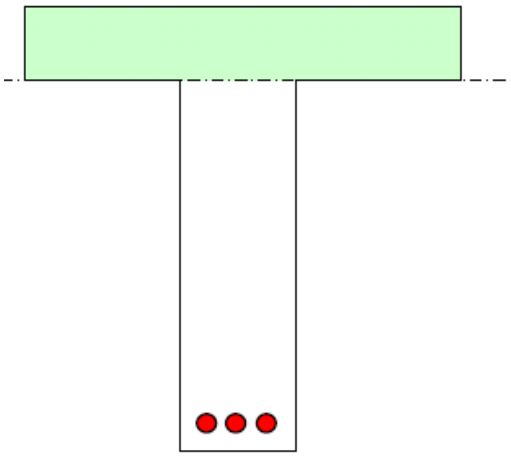
5.5 T形及I形截面梁

2. 两类T形截面的判断

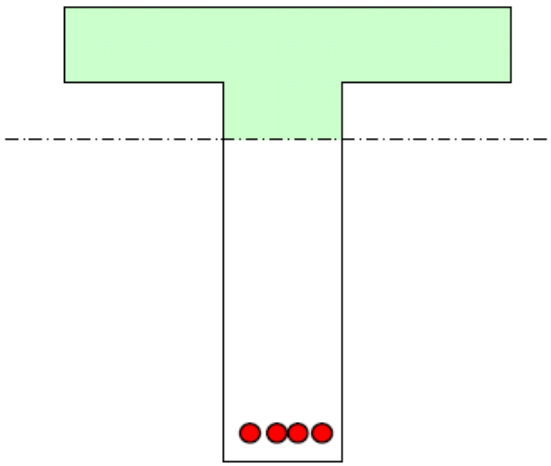
两类T形截面及其判别



$x < h'_f$



$x = h'_f$



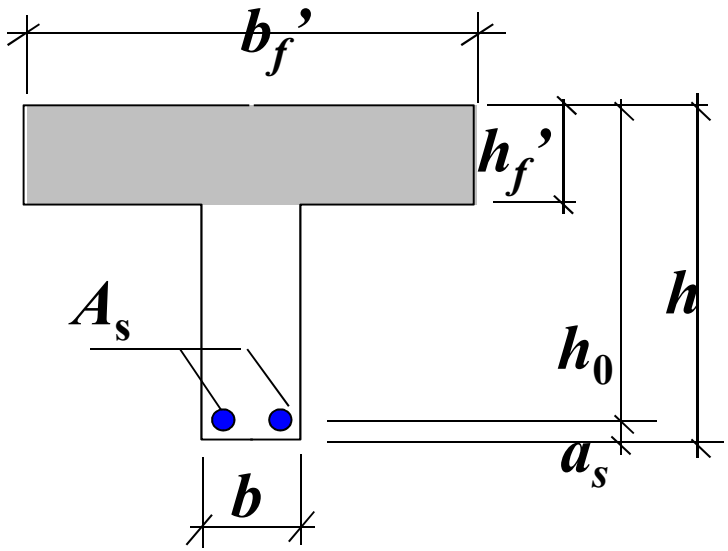
$x > h'_f$

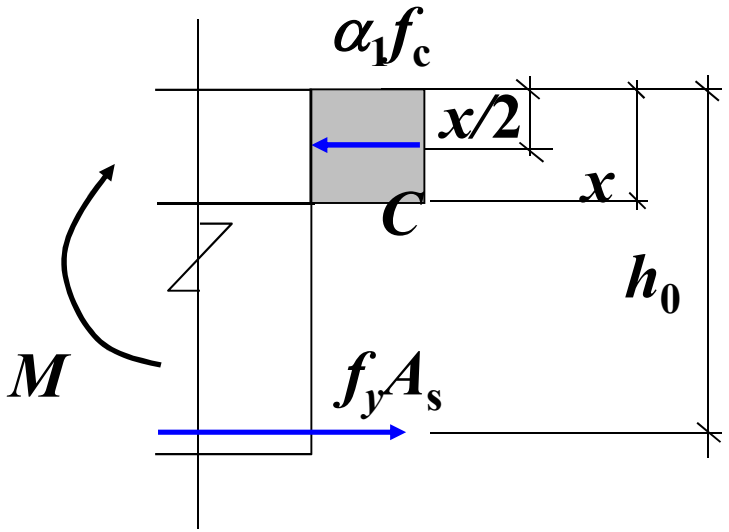
第I类

第II类

5.5 T形及I形截面梁

2. 两类T形截面的判断





$$f_y A_s \le \alpha_1 f_c b_f' h_f', \text{或}$$

$$M \le \alpha_1 f_c b_f' h_f' \left(h_0 - \frac{h_f'}{2}\right)$$

I类

中和轴位于翼缘

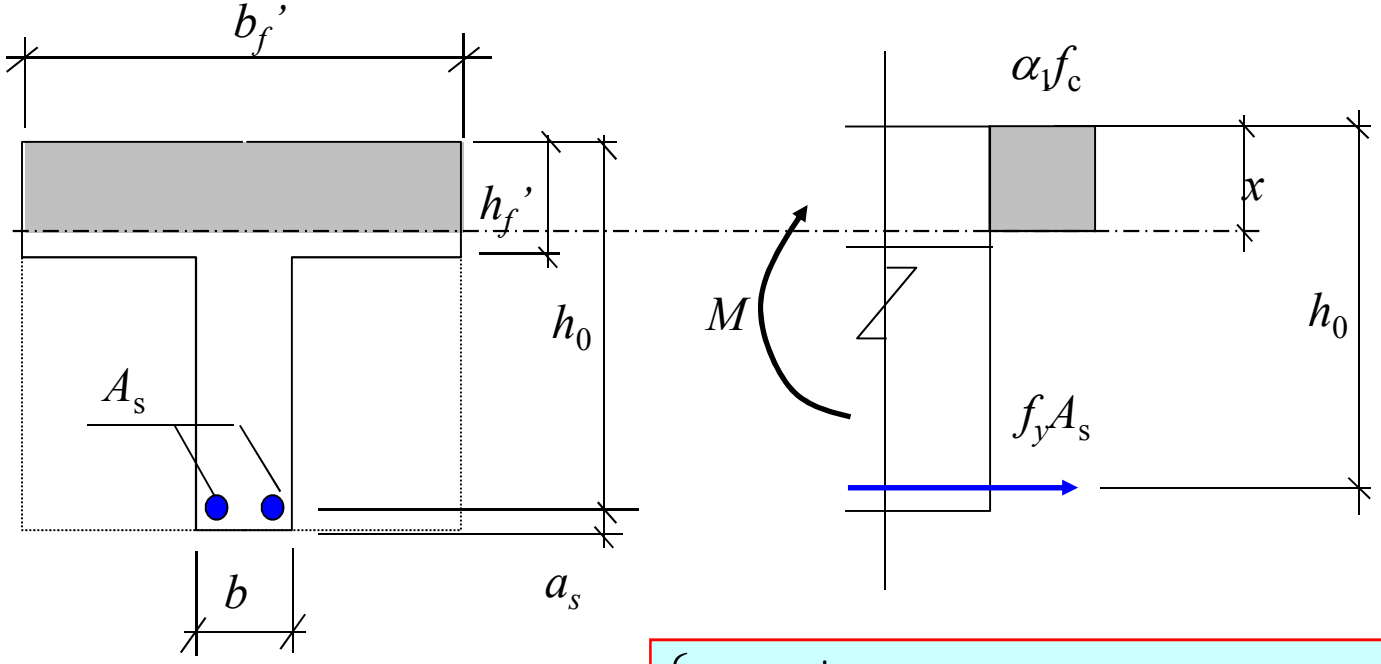
否则

II类

中和轴位于腹板

5.5 T形及I形截面梁

3. 第一类T形截面



按$b_f'\times h$的矩形截面计算

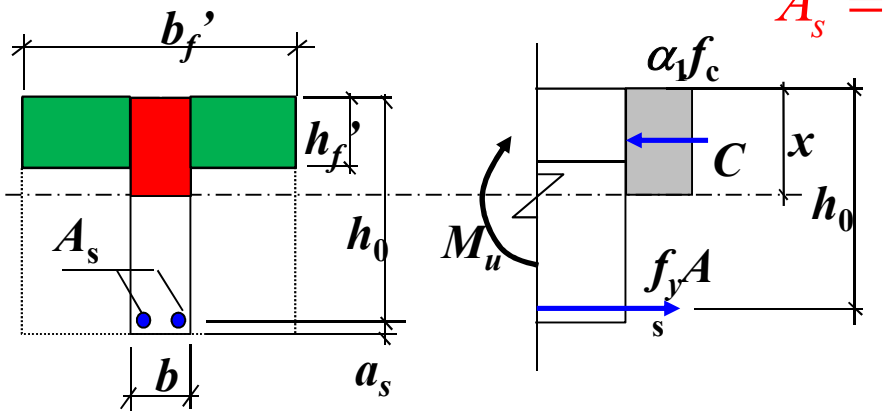
$$\begin{cases}\alpha_1 f_c b_f' x = f_y A_s \\ M_u = \alpha_1 f_c b_f' x(h_0 - \dfrac{x}{2}) = f_y A_s (h_0 - \dfrac{x}{2})\end{cases}$$

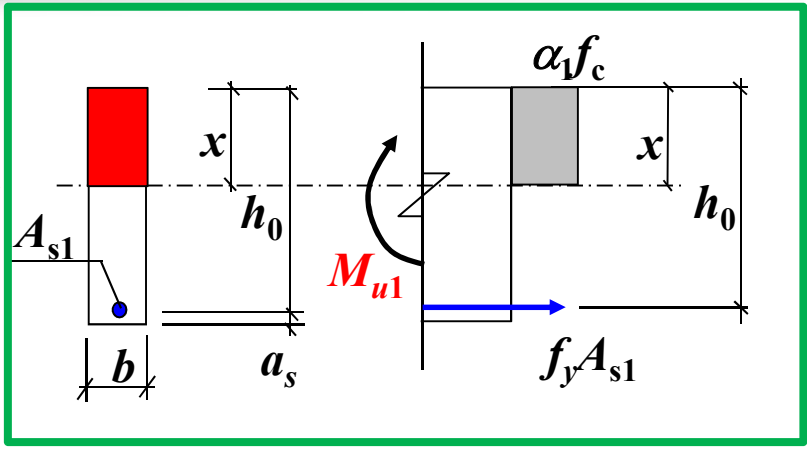
适用条件 $\rho_1 = \dfrac{A_s}{bh} \geq \rho_{\min}$ (非少筋) $\xi \leq \xi_b$ (非超筋，通常可不验算)

5.5 T形及I形截面梁

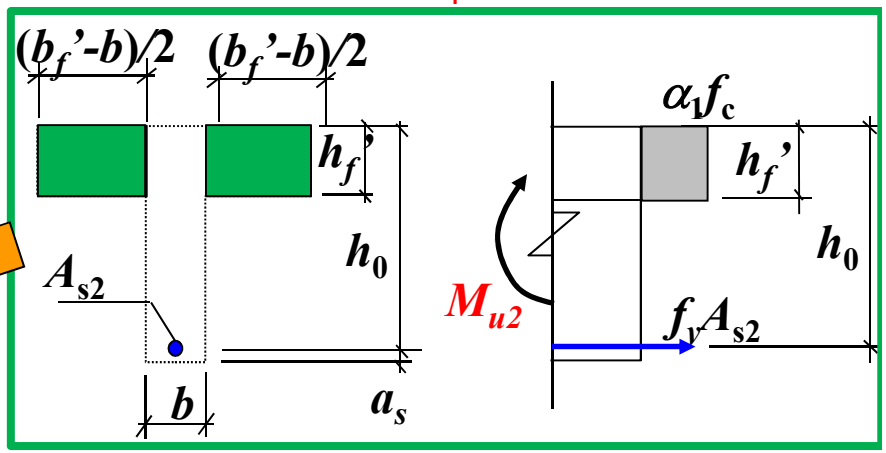
4. 第二类T形截面

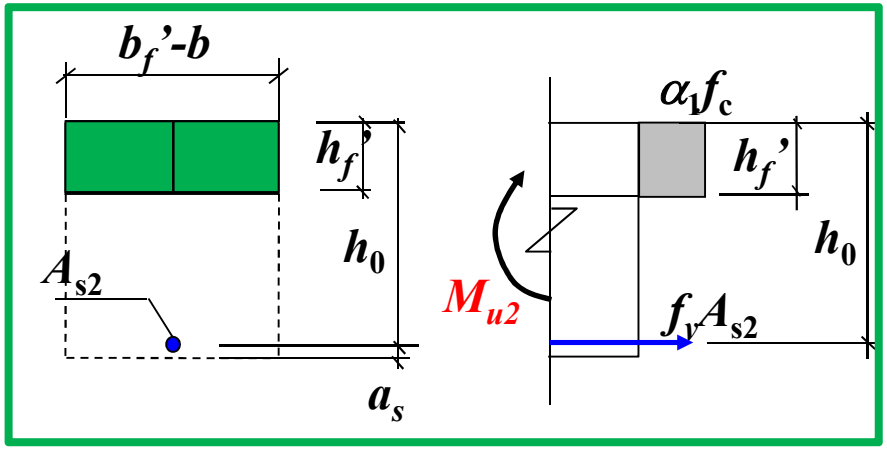
$$A_s = A_{s1} + A_{s2}$$





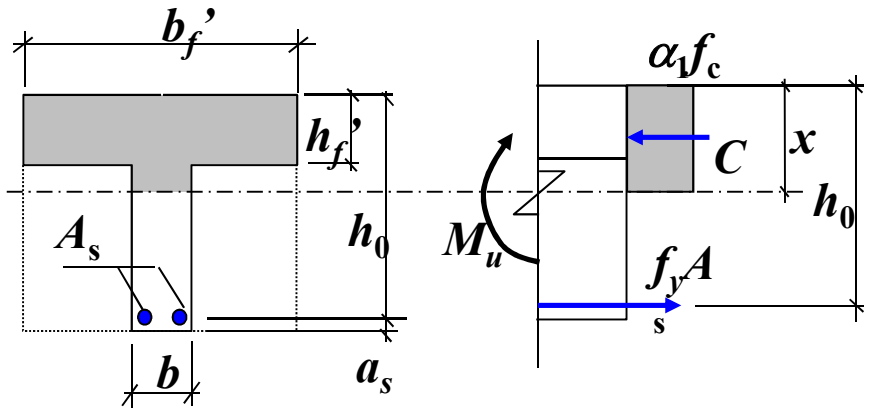
+



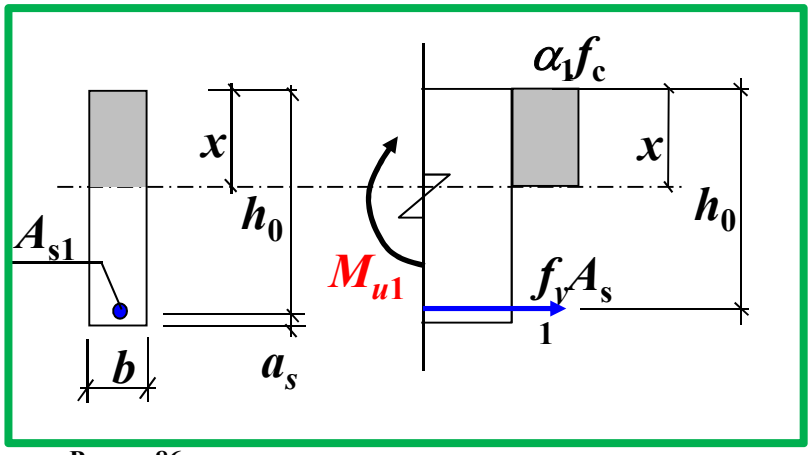


5.5 T形及I形截面梁

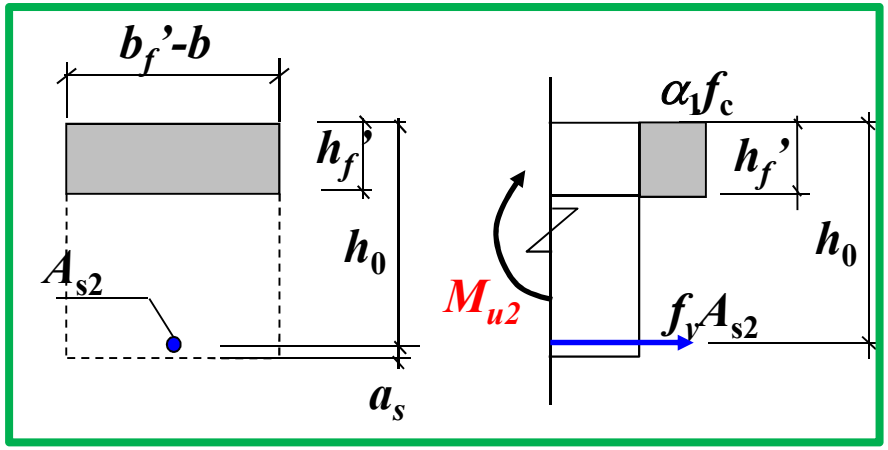
4. 第二类T形截面



$$A_s = A_{s1} + A_{s2}$$

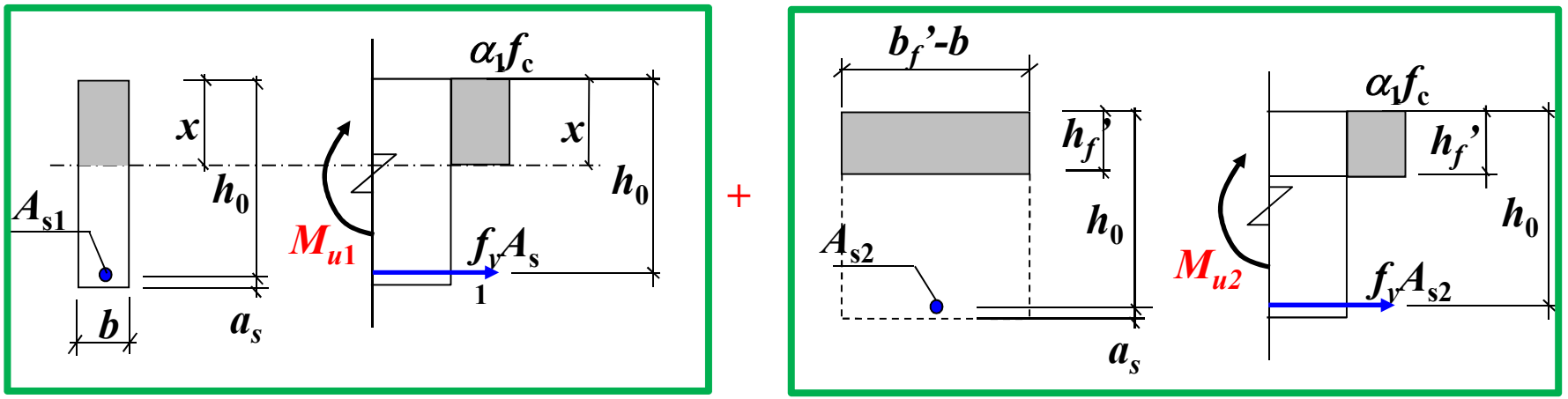


+



5.5 T形及I形截面梁

4. 第二类T形截面



分两部分列方程

$$\sum X = 0 \qquad \alpha_1 f_c b x = f_y A_{s1}$$

$$\alpha_1 f_c (b_f' - b) h_f' = f_y A_{s2}$$

$$\sum M = 0 \qquad M_{u1} = \alpha_1 f_c b x (h_0 - \frac{x}{2})$$

$$M_{u2} = \alpha_1 f_c (b_f' - b) h_f' (h_0 - \frac{h_f'}{2})$$

总方程

$$\sum X = 0 \qquad \alpha_1 f_c b x + \alpha_1 f_c (b_f' - b) h_f' = f_y A_s$$

$$\sum M = 0 \qquad M \le M_u = M_{u1} + M_{u2} = \alpha_1 f_c b x (h_0 - \frac{x}{2}) + \alpha_1 f_c (b_f' - b) h_f' (h_0 - \frac{h_f'}{2})$$

5.5 T形及I形截面梁

5. 承载力计算公式的应用

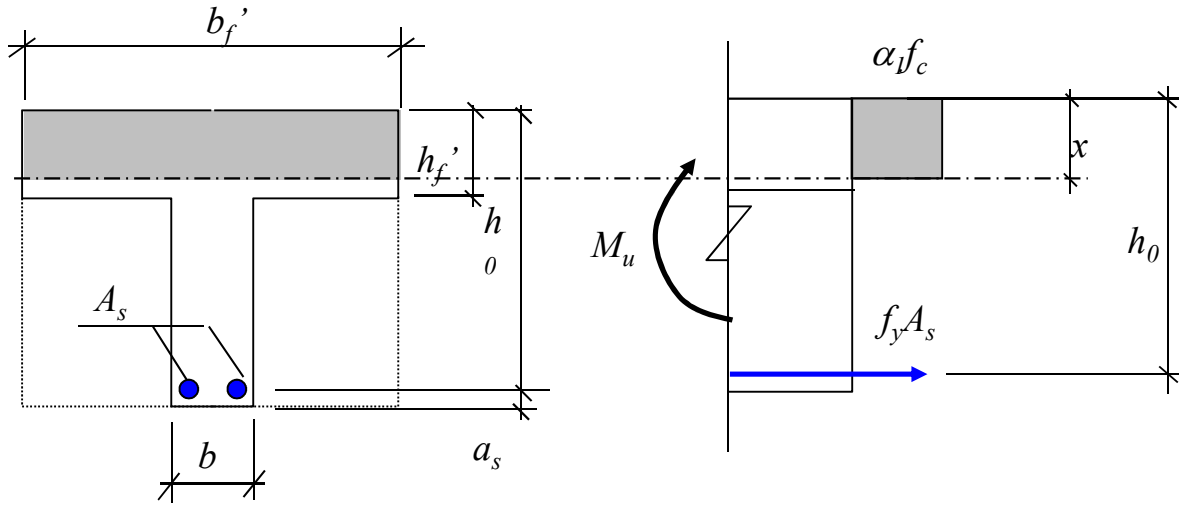
既有构件正截面抗弯承载力

明确b_f'

$$f_y A_s \le \alpha_1 f_c b_f' h_f'$$

I类T形截面

按$b_f' \times h$的矩形截面计算构件的承载力



若$A_s < \rho_{min} bh$

按$b \times h$的矩形截面的开裂弯矩计算构件的承载力

5.5 T形及I形截面梁

5. 承载力计算公式的应用

既有构件正截面抗弯承载力

明确b_f'

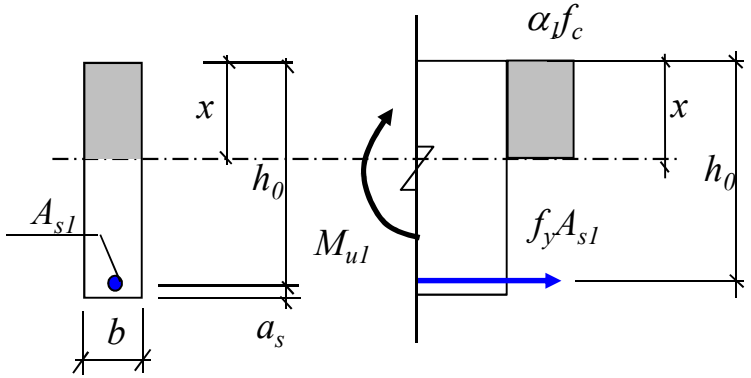
$$f_y A_s > \alpha_1 f_c b_f' h_f'$$

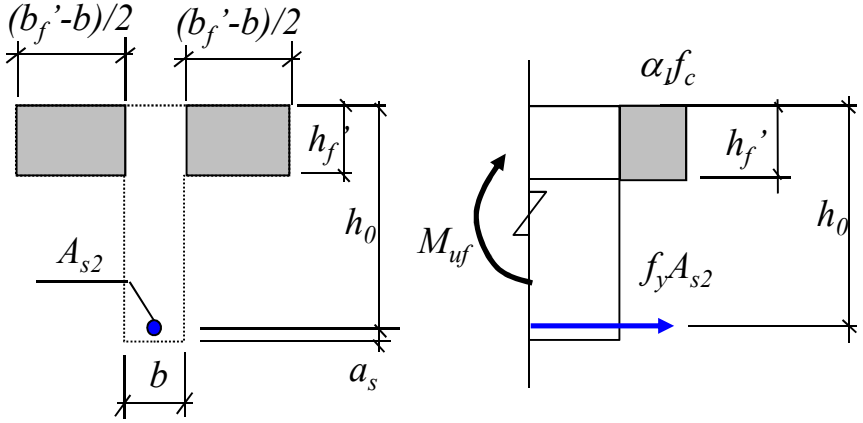
II类T形截面

$$M_{uf}' = \alpha_1 f_c (b_f' - b) h_f' \left(h_0 - \frac{h_f'}{2}\right)$$

按$b \times h$的单筋矩形截面计算M_{u1}

$$M_u = M_{uf}' + M_{u1}$$





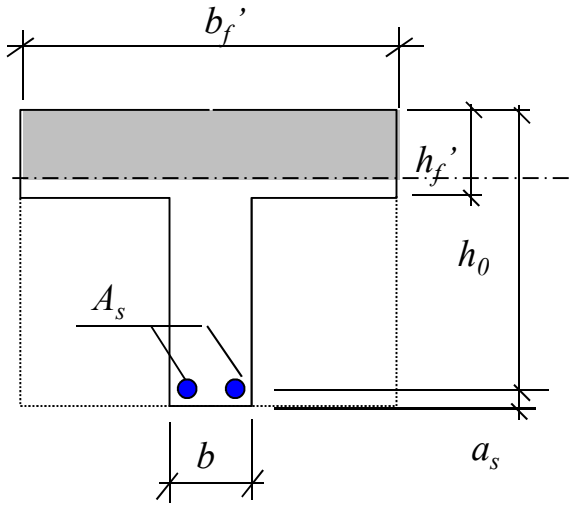
5.5 T形及I形截面梁

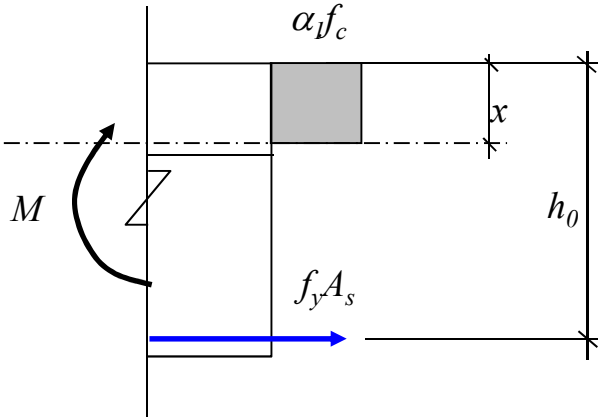
5. 承载力计算公式的应用

基于承载力的截面设计

明确b_f'

$$M \le \alpha_1 f_c b_f' h_f' \left(h_0 - \frac{h_f'}{2}\right)$$





I类T形截面

按b_f'×h单筋矩形截面进行设计

$$\rho = \frac{A_s}{bh} \ge \rho_{\min}$$

5.5 T形及I形截面梁

5. 承载力计算公式的应用

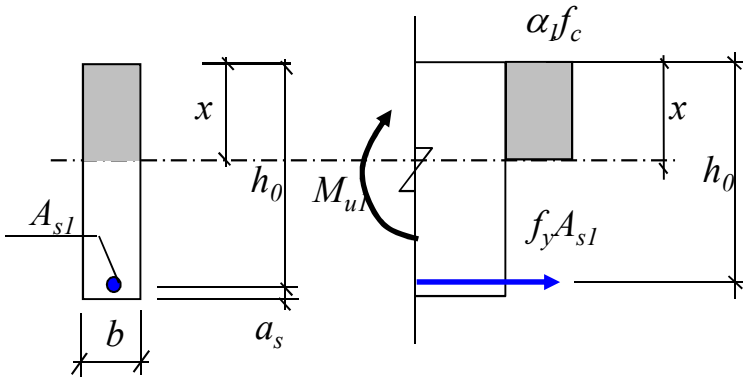
基于承载力的截面设计

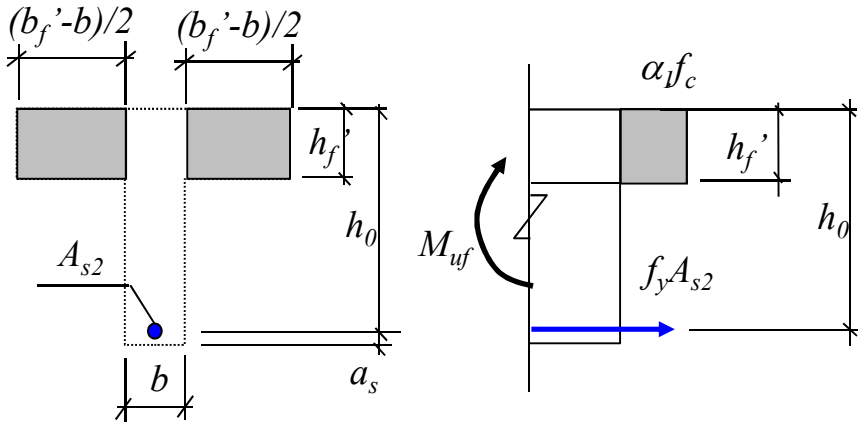
明确b_f'

$$M > \alpha_1 f_c b_f' h_f' (h_0 - \frac{h_f'}{2})$$

II类T形截面

与A_s'已知的b×h双筋矩形截面类似进行设计





5.5 T形及I形截面梁

$$A_{s2}=\frac{\alpha_1 f_c(b_f'-b)h_f'}{f_y} \quad \Leftarrow \quad f_y A_{s2}=\alpha_1 f_c(b_f'-b)h_f'$$

$$M_{u2}=f_y A_{s2}\left(h_0-h_f'/2\right)$$

$$M_{u1}=M-M_{u2}$$

$$\alpha_s=\frac{M_{u1}}{bh_0^2\alpha_1 f_c}$$

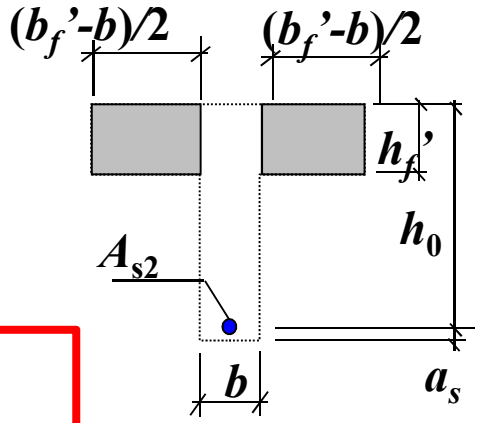
$$\xi=1-\sqrt{1-2\alpha_s}$$

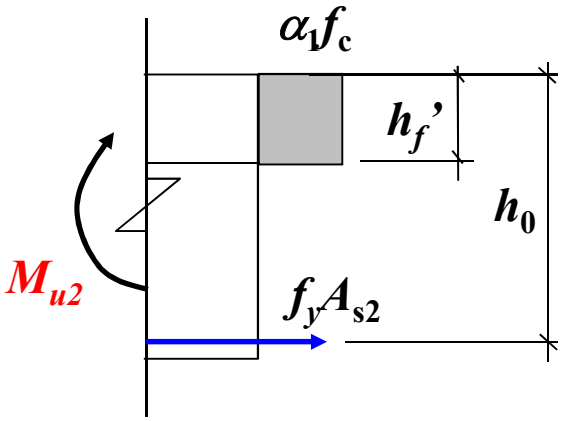
$$A_{s1}=\xi b h_0\frac{\alpha_1 f_c}{f_y}$$

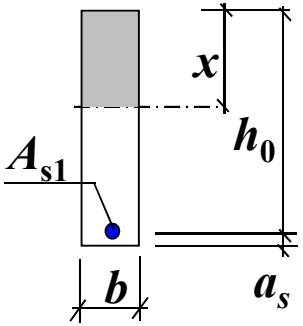
或

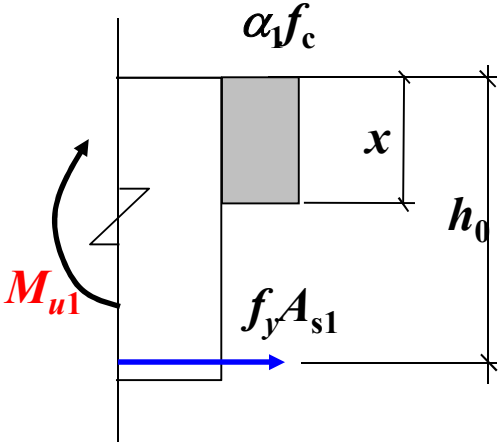
$$\gamma_s=\frac{1+\sqrt{1-2\alpha_s}}{2}$$

$$A_{s1}=\frac{M_{u1}}{\gamma_s h_0 f_y}$$









5.6 公路桥涵工程中受弯构件正截面承载力计算方法

基本假定

（1）构件弯曲后，其截面仍然保持平面；

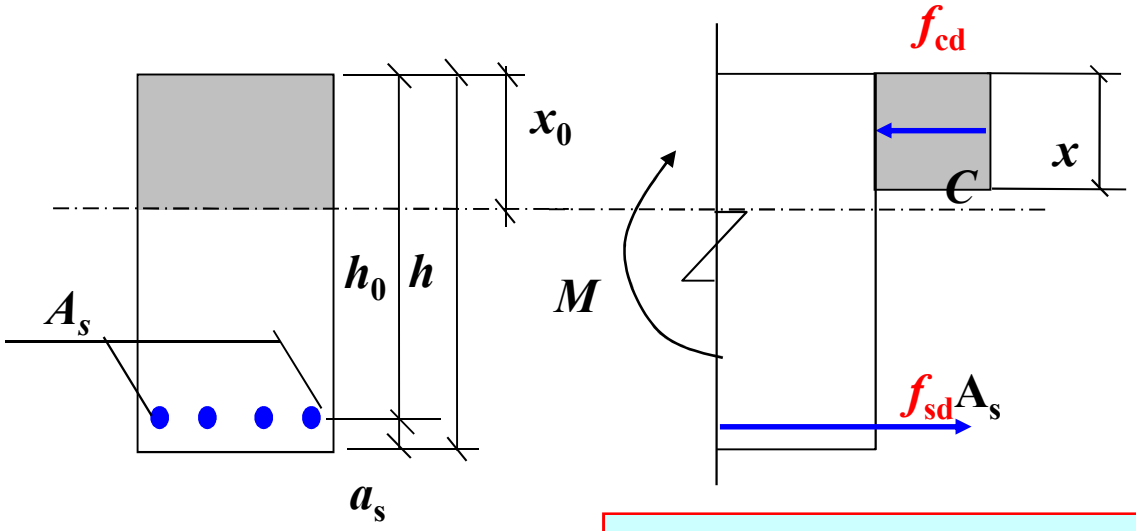
（2）截面受压混凝土的应力图形简化为矩形，其压力强度取混凝土的轴心抗压强度设计值f_{cd}；截面受拉混凝土的抗拉强度不予考虑；

（3）钢筋应力等于钢筋应变与其弹性模量的乘积，但不大于其强度设计值。受拉钢筋的极限拉应变取0.01。

$$\left.\begin{aligned} \sigma_s &= \varepsilon_s E_s \le f_{sd} \\ \sigma_s' &= \varepsilon_s' E_s' \le f_{sd}' \\ \varepsilon_{s,\max} &= 0.01 \end{aligned}\right\}$$

5.6 梁公路桥涵工程中受弯构件正截面承载力计算方法

单筋矩形截面正截面承载力计算



$$\sum X = 0, \quad f_{cd} b x = f_{sd} A_s$$

$$\sum M = 0, \quad \gamma_0 M_d \le f_{cd} b x \left(h_0 - \frac{x}{2} \right)$$

适用条件 $\xi \le \xi_b$ (非超筋)

$A_s \ge \rho_{\min} b h_0$ (非少筋)

5.6 梁公路桥涵工程中受弯构件正截面承载力计算方法

公路桥涵过程与建筑工程关于受弯构件正截面承载力基本计算公式比较

截面类型	建筑工程	公路桥涵工程
单筋矩形	$\alpha_1 f_c b x = f_y A_s$ $M \le \alpha_1 f_c b x \left(h_0 - \frac{x}{2} \right)$	$f_{cd} b x = f_{sd} A_s$ $\gamma_0 M_d \le f_{cd} b x \left(h_0 - \frac{x}{2} \right)$
双筋矩形	$f_y A_s = f_y^{'} A_s^{'} + \alpha_1 f_c b x$ $M \le f_y^{'} A_s^{'} \left(h_0 - a_s^{'} \right) + \alpha_1 f_c b x \left(h_0 - \frac{x}{2} \right)$	$f_{sd} A_s = f_{sd}^{'} A_s^{'} + f_{cd} b x$ $\gamma_0 M_d \le f_{sd}^{'} A_s^{'} \left(h_0 - a_s^{'} \right) + f_{cd} b x \left(h_0 - \frac{x}{2} \right)$
第一类T形	$\alpha_1 f_c b_f^{'} x = f_y A_s$ $M \le \alpha_1 f_c b_f^{'} x (h_0 - \frac{x}{2})$	$f_{cd} b_f^{'} x = f_{sd} A_s$ $\gamma_0 M_d \le f_{cd} b_f^{'} x (h_0 - \frac{x}{2})$
第二类T形	$\alpha_1 f_c b x + \alpha_1 f_c (b_f^{'} - b) h_f^{'} = f_y A_s$ $M \le \alpha_1 f_c b x (h_0 - \frac{x}{2}) + \alpha_1 f_c (b_f^{'} - b) h_f^{'} (h_0 - \frac{h_f^{'}}{2})$	$f_{cd} b x + f_{cd} (b_f^{'} - b) h_f^{'} = f_{sd} A_s$ $M \le f_{cd} b x (h_0 - \frac{x}{2}) + f_{cd} (b_f^{'} - b) h_f^{'} (h_0 - \frac{h_f^{'}}{2})$